

Global Renewable Energy in 2025

How Far Have We Come?

A data-driven overview of renewable energy trends across primary energy and electricity

Source: Our World in Data – Renewable Energy | Authors: Hannah Ritchie, Max Roser, Pablo Rosado

75% of Global GHG Emissions Still Come from Fossil Fuels

75%

of global greenhouse
gas emissions from
fossil fuel burning

5M+

premature deaths
annually from air
pollution

**Since the
Industrial
Revolution**

fossil fuels have
dominated energy
systems worldwide

To reduce CO₂ emissions and air pollution, the world must rapidly shift toward low-carbon energy sources — nuclear and renewable technologies.

Renewable energy includes solar, wind, hydropower, geothermal, bioenergy, wave, and tidal sources. Traditional biomass is typically excluded from modern renewable statistics.

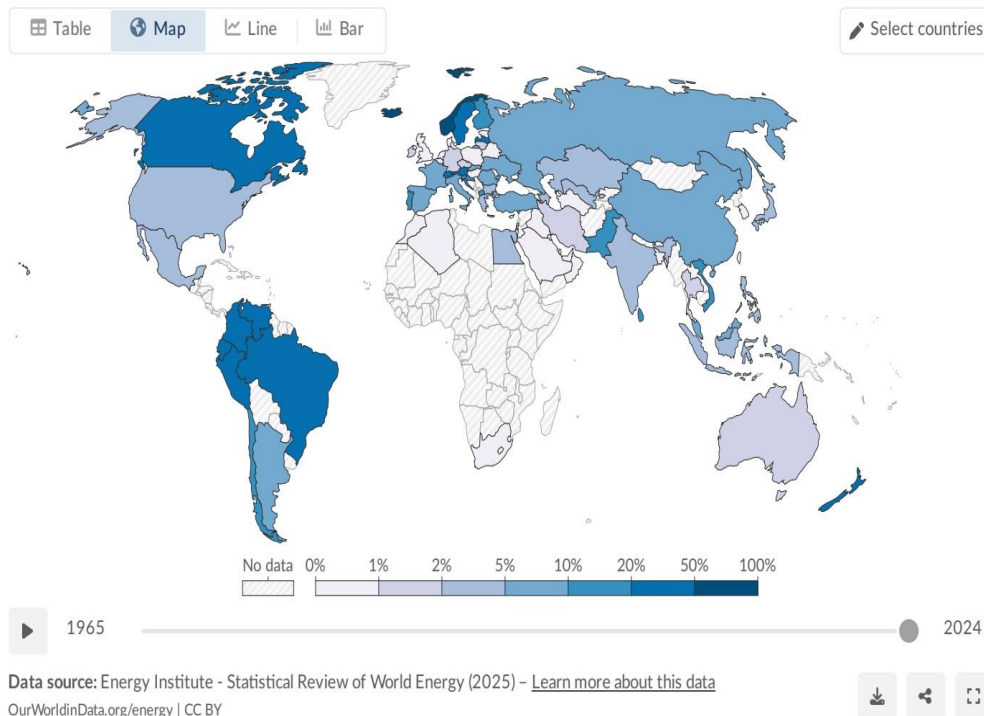
Renewables Now Supply ~14% of Global Primary Energy

~14%

of the world's primary energy now comes from renewable technologies (2024 data, substitution method)

Primary energy encompasses electricity, transport, and heating combined. Because transport and heating are harder to decarbonize, the renewable share in total energy is lower than in electricity alone.

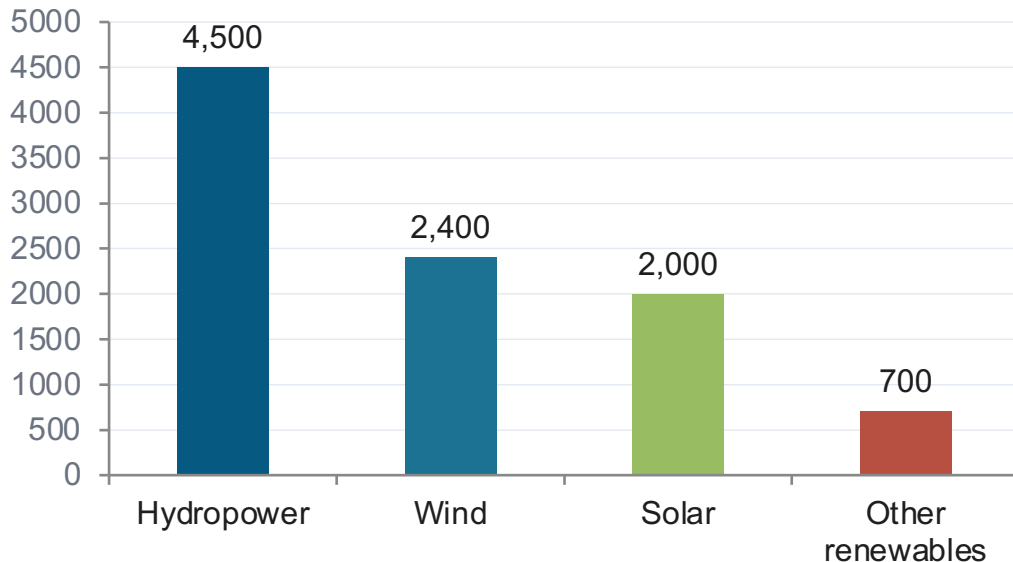
Leaders: Norway, Brazil, New Zealand > 60% |
Middle East & parts of Asia < 5%



Share of primary energy from renewables by country, 2024

Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar

Global Renewable Electricity Generation by Technology
(2024, TWh)



Technology	Generation (TWh)	Share
Hydropower	~4,500	~47%
Wind	~2,400	~25%
Solar	~2,000	~21%
Other renewables	~700	~7%

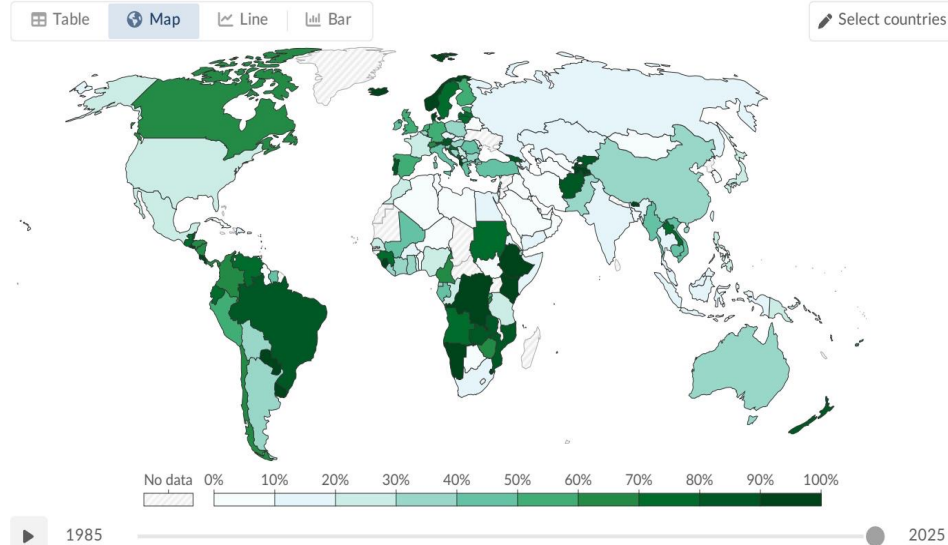
Wind and solar combined now nearly match hydropower's output and are growing much faster. Solar has seen dramatic growth since ~2010, driven by falling costs and policy support.

Almost One-Third of Global Electricity Now Comes from Renewables

Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.

Our World
in Data



Data source: Ember (2026); Energy Institute - Statistical Review of World Energy (2025) - [Learn more about this data](#)
OurWorldinData.org/energy | CC BY

Share of electricity from renewables by country, 2025

Renewable Share of Electricity by Country/Region

Country / Region	Renewable Share
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%

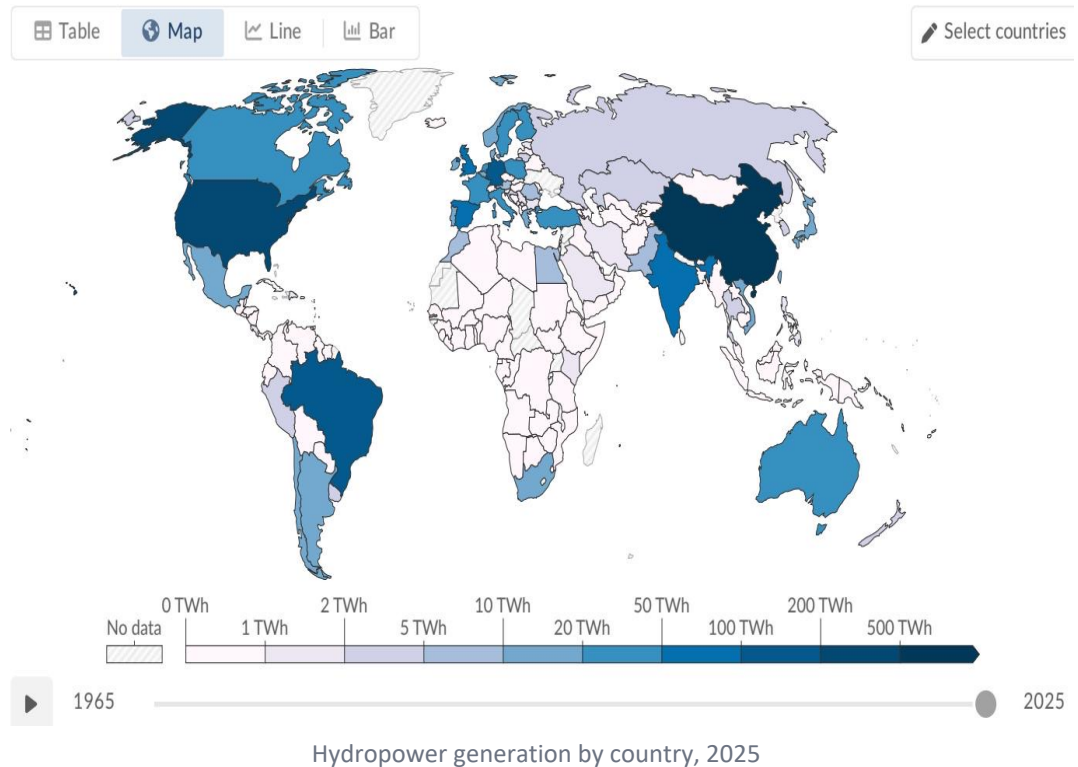
World average: ~30% | Several African and Middle Eastern countries remain below 10%

Hydropower Remains the Largest Renewable Source at ~50% of Generation

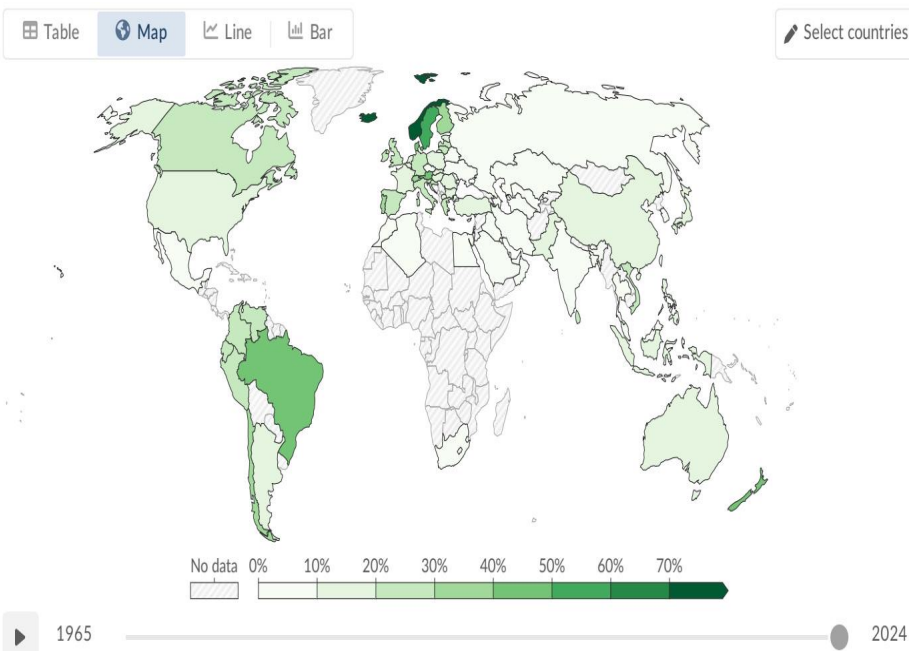
~50%

of all renewable electricity comes from hydropower

Hydropower has been generating electricity at scale for over a century. Top producers include China, Brazil, and Canada.



Wind and Solar Are the Fastest-Growing Energy Technologies in History



Wind power generation by country, 2025

WIND

~2,400 TWh

~25% of renewable electricity | Major in Europe, China, and the United States | Includes onshore and offshore

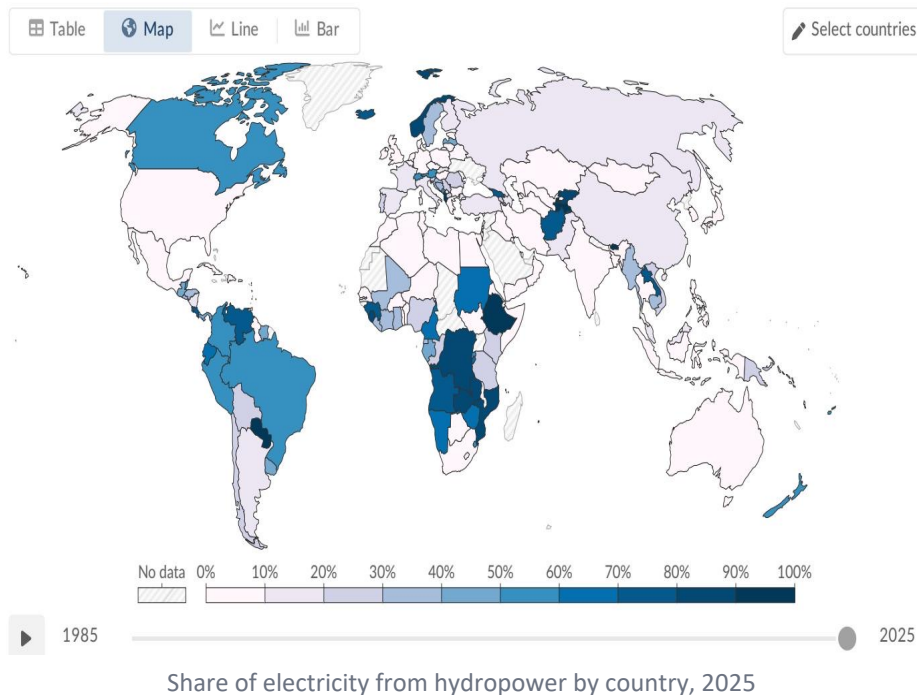
SOLAR

~2,000 TWh

~21% of renewable electricity | Fastest-growing source | Dramatic growth since 2010 driven by falling costs and policy support

Combined, wind + solar now nearly match hydropower output

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America Lead



LEADERS

Scandinavia, Brazil, and Canada benefit from abundant hydropower (65–98% renewable electricity). Denmark and Germany have invested heavily in wind and solar.

LAGGING REGIONS

Much of Sub-Saharan Africa has minimal renewable electricity infrastructure. Many Middle Eastern nations remain almost entirely dependent on fossil fuels for electricity (<10% renewables).

Gap: 98% renewable electricity (Norway) vs. <2% (Saudi Arabia)

Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

01

~14% of primary energy and ~30% of electricity now come from renewables — significant but far from sufficient for climate targets.

02

Hydropower remains dominant at ~50% of renewable electricity, but its growth potential is limited compared to wind and solar.

03

Wind and solar are the fastest-growing energy technologies in history; combined, they now produce nearly as much as hydropower.

04

Stark regional disparities persist: Scandinavia and Latin America exceed 60–90%, while much of the Middle East and Sub-Saharan Africa remain below 10%.

05

Transport and heating lag far behind electricity in the renewable transition, keeping overall primary energy share much lower.

The pace of transition must accelerate dramatically to meet global climate targets.